Scientific Reports

https://doi.org/10.1038/s41598-026-39787-8

Article in Press

Exploring interactions of thermal, acoustic, and aesthetic environments on thermal experience in parks

Received: 16 April 2025

Accepted: 9 February 2026

Published online: 12 February 2026

Cite this article as: Zhang Y., Uchiyama Y. & Sato M. Exploring interactions of thermal, acoustic, and aesthetic environments on thermal experience in parks. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-39787-8

Yu Zhang, Yuta Uchiyama & Masayuki Sato

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

Exploring Interactions of Thermal, Acoustic, and Aesthetic Environments on Thermal Experience in Parks

Yu Zhang[1], **Yuta Uchiyama**[1], **and Masayuki Sato**[1,2*]

[1]Graduate School of Human Development and Environment, Kobe University, 3 11 Tsurukabuto, Nada-ku 657 8501, Japan
[2]Advanced Research Institute for Well-being, Kobe University, 3 11 Tsurukabuto, Nada-ku 657 8501, Japan
[*]msat@port.kobe-u.ac.jp

ABSTRACT

The thermal comfort of urban parks is crucial for residents' well-being and urban livability. There is limited research on the integration of different environmental factors in urban parks, with most studies focusing on hot, sunny days in summer. This study explored the interactions of multi-sensory factors, including thermal, acoustic, and aesthetic, on thermal experience in parks across various weather types. The results reveal that thermal conditions directly dominate thermal experience. Acoustic comfort has a relatively weak positive effect on thermal comfort, and this effect disappears under strong heat-stress conditions. Natural sounds help increase thermal acceptability, whereas mechanically dominated soundscapes have the opposite effect. Aesthetic satisfaction also contributes to higher thermal acceptability. Compared with blue spaces, aesthetic satisfaction with green spaces exerts a more consistently positive influence on thermal experience. The negative effect of PET is significantly weaker than the positive effects of perceived acoustic comfort and aesthetic satisfaction on overall satisfaction. The revealed multi-sensory interactions yield practical recommendations for urban planners and designers in addressing climate change and enhancing urban livability.

Keywords: Outdoor Thermal Comfort , Landscape Aesthetic Quality , Acoustic Environment , Microclimate , Green-Blue Spaces

ARTICLE IN PRESS

Abbreviations

Ta	Air temperature (°C)	**RH**	Relative humidity (%)
Va	Wind speed (m/s)	**Tmrt**	Mean radiant temperature (°C)
Tg	Globe temperature (°C)	**G**	Global radiation (W/m^2)
PET	Physiological equivalent temperature (°C)	**TSV**	Thermal sensation vote
TCV	Thermal comfort vote	**TAV**	Thermal acceptability vote
LAeq	A-weighted equivalent continuous sound level	**AcSV**	Acoustic sensation vote
AcCV	Acoustic comfort vote	**AeSV**	Aesthetic satisfaction vote
AeSVG	Aesthetic satisfaction vote for green space	**AeSVB**	Aesthetic satisfaction vote for blue space
OSV	Overall satisfaction vote		

Introduction

Climate change and the urban heat island effect have significantly intensified heat stress for urban residents[1,2]. Outdoor spaces, as essential extensions of urban living environments, play a critical role in supporting daily activities[3]. In recent years, improving outdoor thermal environments to enhance residents' thermal comfort has become a growing research focus. Among outdoor spaces, parks serve as vital components of the urban ecosystem, offering health benefits and being highly regarded by residents[4]. Research indicates that urban parks are capable of mitigating air and noise pollution, improving thermal conditions, and alleviating stress, which enhance residents' physical and mental well-being[5]. Thus, improving the thermal environment of urban parks is crucial for enhancing residents' thermal comfort, which in turn promotes urban livability and supports sustainable development.

ARTICLE IN PRESS

Outdoor thermal comfort (OTC), an important indicator for assessing the interaction between the thermal environment and individual comfort, is evaluated through objective measurements and subjective surveys[6]. Numerous studies have demonstrated that meteorological factors, particularly air temperature, relative humidity, wind speed, and solar radiation, are key variables influencing thermal comfort in urban open spaces[7]. Among these factors, air temperature not only directly affects individuals' thermal perception but also indirectly influences their spatial behavior, moderated by such factors as activity type, seasonal variation, and weekend effects[8]. However, relying solely on meteorological parameters cannot fully capture the complexity of OTC[9]. Studies indicate that microclimate variables explain only about 50% of the variation in thermal perception, and psychological adaptability is considered a key factor contributing to the discrepancy between subjective thermal perception and objective measurements[10]. Researchers increasingly focus on how sensory factors, acoustic environment, visual landscape, and psychological state regulate thermal comfort, emphasizing their interaction with the thermal environment as key to achieving true comfort.

In some laboratory settings, multi-domain research on environmental perception focuses on the interaction between at least two sensory dimensions[11], such as heat-sound, heat-visual, heat-air quality, and sound-visual interactions. However, the conclusions of such studies often lack consistency and are subject to certain controversies[12]. Nonetheless, laboratory research has provided a theoretical basis for identifying and verifying relevant mechanisms, and has laid an important foundation for the study of multi-sensory environmental perception in outdoor settings. However, similar issues exist in research on urban outdoor thermal environments. Therefore, further exploration of the interaction mechanisms among multiple sensory factors in urban open spaces is crucial for advancing thermal comfort theory and informing comprehensive urban spatial optimization.

Acoustic Environment and Outdoor Thermal Comfort

The acoustic environment, an important factor influencing OTC, remains controversial regarding its mechanisms and specific impacts. Previous studies have shown that various aspects of the acoustic environment, such as sound pressure level, source type, perceived sound level, and acoustic comfort, can influence an individual's thermal perception. For instance, Tsai and Lin found that higher equivalent continuous sound levels (LAeq) are often observed in neutral thermal environments, whereas lower sound levels tend to occur in warm or hot conditions[13], suggesting a correlation between sound level and thermal environment. However, this relationship is neither unidirectional nor stable, as other studies have revealed a complex interaction between the acoustic environment and thermal perception. Some findings suggest that individuals experience stronger thermal sensation in quieter environments than in noisy urban settings[14]. By contrast, other studies indicate that increased noise levels may slightly enhance thermal perception[15]. Some studies have indicated that the impact of noise on thermal comfort becomes more pronounced under higher temperature conditions[16,17]. Furthermore, the effect of sound on alleviating thermal stress is moderated by humidity, with significant reduction observed when relative humidity ranges from 25% to 70%[18]. However, accounting for landscape aesthetic satisfaction, acoustic quality does not significantly influence thermal perception[19]. Some studies have reported no statistically significant relationship between sound (or noise) and thermal comfort[19–21]. The influence of sound type and intensity on thermal comfort varies across studies. Notably, thermal comfort significantly decreases during summer in environments with intense traffic noise[20,22].

An environmental chamber experiment confirmed a complex interaction between sound pressure level and temperature. At moderate levels, thermal comfort votes, overall comfort, and environmental comfort are jointly influenced by both factors. Under extreme conditions, one factor dominates. The combined thermal-acoustic effect threshold is more lenient with music but stricter with fan noise[23]. Current research indicates a complex and variable relationship between thermal comfort and the acoustic environment, influenced by climatic conditions, individual differences, and multiple interacting factors. Thus, further investigation into how the acoustic environment affects OTC remains a crucial issue in this field.

Visual Environment and Outdoor Thermal Comfort

In a multi-sensory environment, visual perception, a key environmental stimulus, significantly influences an individual's thermal comfort experience. It includes not only physical attributes, such as brightness and color, but also subjective evaluations of landscape aesthetics and the emotional responses they evoke. The "hue-heat hypothesis" suggests that people associate certain colors with temperature, categorizing reds and oranges as "warm colors," and blues and greens as "cool colors." This psychological association can influence an individual's perception of heat to some extent. Many studies explore the impact of visual factors, such as illuminance and the color of lights or objects, on thermal sensation in indoor environments, both in laboratory and field settings[24–26]. Outdoor studies have found that under moderate lighting, individuals exhibit greater thermal sensitivity than under strong or dim lighting conditions, suggesting that moderate light may enhance heat perception. In winter outdoor environments, cool-colored backgrounds tend to evoke stronger cold sensations, supporting the hue-heat hypothesis. However, this effect is context-dependent in summer, showing statistical significance only under low heat stress conditions[27]. Currently, research findings on the actual impact of light color on thermal comfort remain inconclusive[12]. Huebner et al. found higher thermal comfort and acceptability under 2700 K lighting than under 6500 K lighting[28], while Candas and Dufour reported the opposite result[29].

ARTICLE IN PRESS

Studies show that visual landscapes significantly affect thermal comfort and overall well-being, and enhancing visual comfort can help reduce thermal discomfort[30,31].The primary factors for assessing indoor visual comfort are luminosity-related parameters, such as horizontal and vertical illuminance, scene or spot illuminance, solar irradiance, spectral composition, color temperature correlation, and surface optical properties[32–34]. Outdoor visual settings are inherently more complex and dynamic than indoor environments are. Natural elements, such as trees, water, and grass, coexist with human-made structures including roads, buildings, and billboards, while such factors as lighting, color, spatial layout, and visual openness are constantly changing[35]. Additionally, outdoor environments are frequently influenced by dynamic disturbances, such as occlusion, reflections, and pedestrian movement. The variability in color, light, and scenery is largely driven by the temporal fluctuations of natural light spectra, further increasing the complexity of visual stimuli[36,37]. Given the limited duration of individual exposure outdoors, people often find it challenging to form clear and stable perceptions of single visual variables amid such multi-sourced and multidimensional stimuli. To address this limitation, researchers have increasingly integrated overall visual aesthetic appeal into thermal comfort frameworks, employing aesthetic satisfaction as a proxy for outdoor visual comfort[38,39]. This indicator not only reflects an individual's overall subjective satisfaction with the visual environment but also encompasses emotional states and psychological adaptation, thereby providing a more comprehensive measure of how visual experience influences thermal perception.

Natural landscapes, particularly green and blue spaces, help individuals reduce stress, anxiety, and tension, while enhancing thermal acceptability and overall comfort[40,41]. Such elements as green spaces, water bodies, and shade are generally linked to improved thermal comfort. Notably, tree canopy coverage and height are recognized as key factors in regulating the thermal environment[42,43]. In urban parks, blue-green spaces constitute a vital component, differing in visual characteristics and psychological associations. Blue spaces, such as water bodies, are often linked to perceptions of coolness, fluidity, and openness, whereas green spaces, like vegetation, evoke feelings of shelter, vitality, and security. Different combinations of blue and green landscapes elicit varying emotional responses and levels of visual satisfaction, which may indirectly affect an individual's thermal perception through emotional regulation mechanism[44]. Current research on visual environment and thermal comfort mainly addresses overall aesthetic perception, with limited focus on aesthetic satisfaction of blue and green spaces. Thus, separately examining public evaluations of these landscapes may better reveal how visual elements specifically influence thermal comfort regulation.

This study addressed limitations in existing research by selecting two different types of parks to evaluate OTC from a multi-sensory perspective. The objectives were to: (1) examine differences in thermal, acoustic, and aesthetic comfort in two parks and assess how respondent attributes influence thermal comfort; (2) explore the independent effects of the thermal environment, acoustics, and aesthetics on thermal comfort; and (3) evaluate the combined influence of thermal, acoustic, and aesthetic factors on overall thermal comfort. Our findings highlight how thermal, acoustic, and aesthetic factors influence thermal comfort, providing a theoretical foundation for park planning and management that prioritizes residents' well-being. These results not only deepen our understanding of urban thermal comfort but also offer practical guidance for improving urban thermal environments and advancing sustainable urban development.

ARTICLE IN PRESS

Methodology

Study Site

This study was conducted in Kobe and Osaka City, Japan. Both cities are located in a Cfa climate zone, which is classified as a humid subtropical climate with hot summers in the Köppen climate classification. Both cities are located near the coast; a key difference between them is that Kobe is bordered by the Rokko Mountains, while Osaka is relatively flat. In both cities, rainfall and humidity peak in the summer, intensifying the perception of heat. August marked the hottest month in nearly a decade, with average, maximum, and minimum temperatures recorded at 28.88°C, 32.44°C, and 26.42°C in Kobe, and 29.21°C, 33.9°C, and 26.01°C in Osaka, respectively. High temperatures and humidity in the summer have resulted in frequent occurrences of heat-related illnesses, presenting challenges to residents' thermal comfort.

Two parks were selected for the field research: East Park in Kobe and Nagai Park in Osaka. East Park, located in the Sannomiya district, the city center of Kobe, is a restored space covering an area of 2.7 ha, featuring a lawn, square, fountain, and other landscape elements, and occasionally hosts music events and sales. Nagai Park in Osaka City, a large comprehensive park with an area of 66.3 ha, features a stadium, gymnasium, running track, and a botanical garden with a natural lake. Because of their unique characteristics, both parks serve as valuable settings for studying OTC, with three measurement points in East Park and four in Nagai Park. The observation points (A–G) were selected within the two parks, and their locations are summarized in figure (Figure 1).

The observation points in this study were selected to represent a variety of typical park spaces, including open areas, such as lawns and plazas, as well as semi-open spaces, like shaded forest areas. Specifically, two observation points were located near natural lakes, representing waterfront spaces, while two others featured artificial water features, that is, ponds or fountains. The main landscape features of each observation point are shown in table 1.

ARTICLE IN PRESS



Figure 1. Study Area and Observation Points. The location map was obtained from Google Earth Pro (version 7.3; https://www.google.com/intl/en/earth/versions/) and annotated by the authors to indicate the observation points (A–G) and to prepare the final layout. The on-site photographs taken by the authors illustrate the site conditions. Images/Maps data: Google Earth; Maxar Technologies; CNES/Airbus.

ARTICLE IN PRESS

Experimental Design

The field study was conducted simultaneously at East Park and Nagai Park from August 18 to 24, 2024. Data collection began after 9:00 AM and ended before 5:00 PM to fit park operating hours. The questionnaire survey and environmental measurements were conducted concurrently, with the distribution of questionnaires restricted to a 5-meter radius around the equipment. The instruments were placed in locations that did not disrupt pedestrian flow, and participants were randomly selected from passersby to complete their questionnaires under stable conditions. All questionnaire responses were collected anonymously, because they included private information, such as health and physical conditions.

Field Measurements

The environmental parameters measured included air temperature, RH, global radiation, wind speed, globe temperature, and sound levels. A 0.75-meter diameter black globe was used to measure globe temperature, with the temperature sensor fixed inside. All measurement devices were installed 1.5 meters above the ground, automatically recording data every five minutes. The sound level sensor was sourced from SL-4023SD, while the remaining environmental sensors and the data logger were provided by the HOBO series. A summary of the measurement equipment and related details is provided (see the Appendix). The mean radiant temperature (Tmrt) was computed using the following equation (1)[45].

$$T_{mrt} = \left[(T_g + 273.15)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{0.25} - 273.15 \quad (1)$$

where ε denotes the emissivity of the black globe (with $\varepsilon = 0.95$ in this study), and D represents the diameter of the black globe (with D = 0.075, in this study).

ARTICLE IN PRESS

Table 1. Landscape Features in Observation Points

Park	Observation Point	Landscape Features
Nagai Park	Point A	The lakeside forest understory primarily consists of broadleaf trees and exposed soil.
	Point B	The island in the natural lake has a central concrete-paved road, surrounded by a tree-shrub-grass community.
	Point C	Adjacent to the park's main road, with broadleaf trees and scattered groundcover.
	Point D	Near the city's main road; paved hard surfaces, with nearby ponds and planted trees.
East Park	Point E	Paved hard surfaces, with nearby fountains and scattered trees.
	Point F	Resting area primarily composed of tree pits, with brick-dominant hardscaping.
	Point G	Lawn area with scattered trees along the perimeter.

Questionnaire survey

The questionnaire consisted of three sections, with the first focusing on personal characteristics, including gender, age, activity type, clothing insulation. Clothing insulation and metabolic rate values are specifically determined based on the guidelines provided in ASHRAE Standard 55 and ISO 7730[46,47]. The second section primarily assessed subjective perceptions on thermal, acoustic, and aesthetic experience, as well as preferences regarding landscape and policies. In this context, the thermal sensation vote (TSV), thermal comfort vote (TCV), and thermal acceptability vote (TAV) were measured using a 7-point scale, whereas the acoustic sensation vote (AcSV), acoustic comfort vote (AcCV), aesthetic satisfaction vote (AeSV) with blue-green spaces, overall satisfaction vote (OSV), and the perceived urgency of policies for improving the thermal environment were assessed using a 5-point voting scale. The third section explores the characteristics of respondents' park visits, including the frequency of visits, the time required to reach the park, modes of transportation, and whether they were accompanied. The detailed questionnaire is provided in the Appendix.

Thermal Indexes

Physiological equivalent temperature (PET), predicted mean vote (PMV), universal thermal climate index (UTCI), and standard effective temperature (SET) are commonly employed in thermal comfort assessments[48]. In our study, PET was used as the thermal comfort indicator. PET is derived from the Munich energy-balance model for individuals (MEMI), which physiologically models the human body's thermal needs[6]. PET has gained increasing prominence in recent research on OTC owing to its ability to enable comparative assessments of thermal comfort requirements across diverse environments[49]. PET values for all respondents were calculated using the RayMan model throughout the study period, based on the default setup in the software, which includes standard parameters, such as a height of 1.75 meters and weight of 70 kilograms for a typical man, along with typical clothing insulation and activity levels.

Acoustic and Aesthetic Indexes

Sound level intensity is a fundamental indicator for assessing acoustic features. The A-weighted LAeq (LAeq, dBA) was used to evaluate noise levels, and the formula for its calculation is given in Equation (2).

$$L_{Aeq} = 10\log\frac{1}{T}\int_0^T \left(\frac{P(t)}{P_0}\right)^2 dt \tag{2}$$

Here, L_{Aeq} denotes A-weighted equivalent continuous sound level (dBA); T is the total measurement duration; $P(t)$ represents the instantaneous sound pressure at time t; and P_0 is the reference sound pressure (20 μPa) used in this study.
In this study, The LAeq values were directly measured using a sound level meter, which automatically calculates the A-weighted equivalent continuous sound level over the specified five-minute intervals. Additionally, subjective data were collected through a questionnaire, which captured respondents' perceptions of the sound environment. Specifically, the survey assessed the perceived intensity of three types of sounds—natural, artificial, and mechanical—as well as the respondents' overall satisfaction with the sound environment. The aesthetic indicators focus on respondents' satisfaction with the landscape's overall aesthetics, as well as with blue and green spaces individually, reflecting the visual appeal of the environment.

Ethics Declarations

The questionnaire used in this study was reviewed and approved by the Research Ethics Review Committee of the Kobe University Graduate School of Human Development and Environment. The study was conducted in accordance with relevant ethical guidelines and regulations. Oral informed consent was obtained from all participants prior to their participation in the questionnaire survey. Participants completed the questionnaire anonymously and were free to skip any questions they did not wish to answer.

ARTICLE IN PRESS

Table 2. Environmental Attributes across Observation Points

Observation Point	Solar Radiation (W/m^2)	Air Temperature (°C)	Globe Temperature (°C)	RH (%)	Mean Wind Speed (m/s)
A	121.24	33.57	34.60	51.84	0.66
B	615.67	34.22	41.04	55.47	0.84
C	79.58	33.05	34.80	54.98	0.75
D	686.96	35.49	43.75	44.82	0.58
E	636.42	33.51	42.12	62.17	0.66
F	166.00	34.08	36.86	60.75	0.31
G	439.31	31.79	38.12	68.93	0.55

Data Analysis

We first conducted descriptive statistical analysis to present the respondents' demographic characteristics and the distribution of thermal perception. Second, to visually demonstrate the independent effects of the thermal, acoustic, and aesthetic environments on thermal perception, single-variable regression analyses were performed using the binned data. Specifically, PET values were grouped into 1°C intervals, a method commonly used in previous studies to reduce data variability and to better visualize trends in thermal comfort studies. Subsequently, controlling for potential confounding factors, such as park type, gender, age, and visit characteristics, an ordered logistic regression model was employed to analyze the interactions between thermal perception and the acoustic and aesthetic environments. Finally, the ordered logistic regression model was developed to examine the relationships between thermal, acoustic, aesthetic and overall satisfaction.

Results

Descriptive Results

Respondents' attribute and visit characteristics

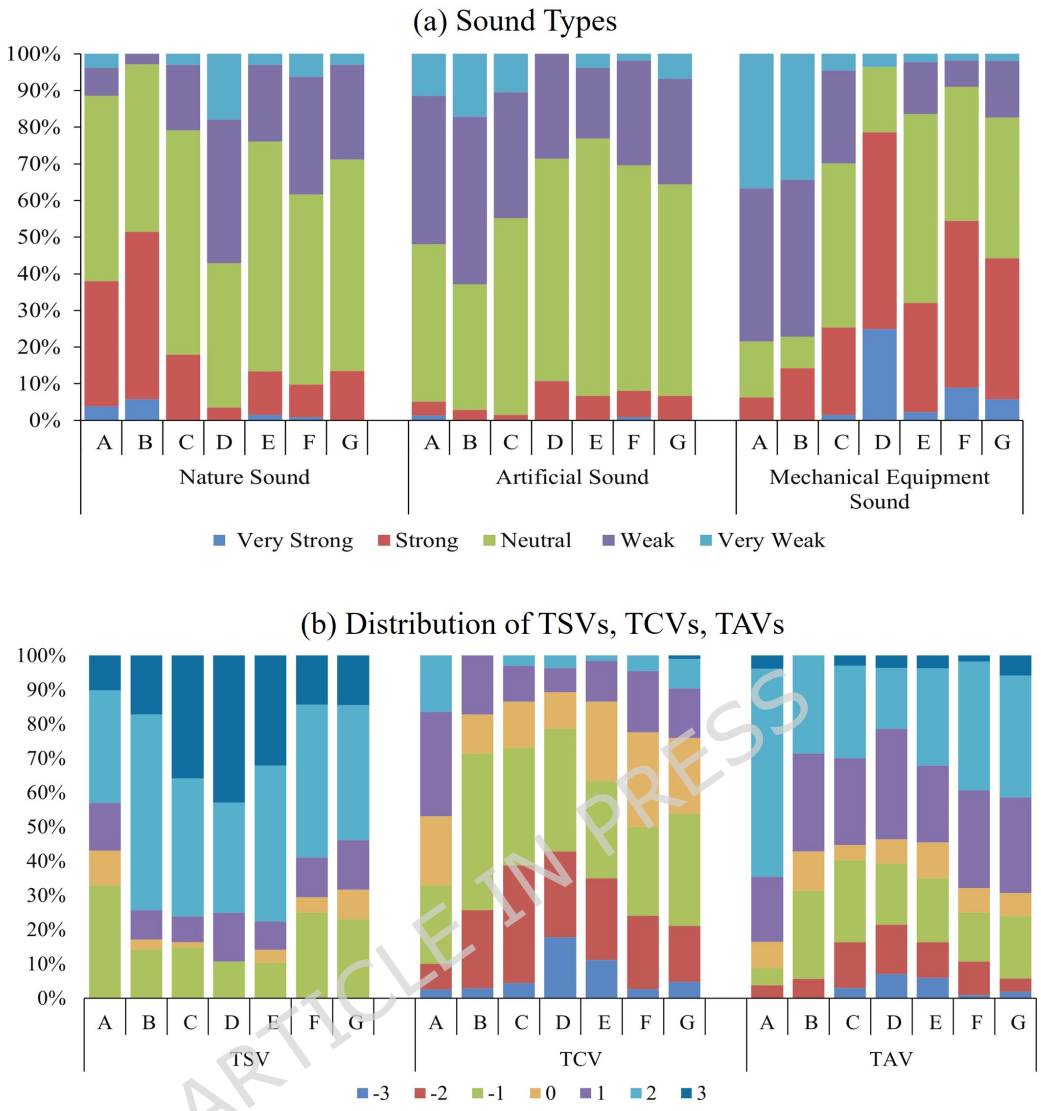
A total of 559 valid questionnaires were gathered, with 209 from Nagai Park and 350 from East Park (see the Appendix). Among the respondents, 415 (74.24%) participated on sunny days, while 144 (25.76%) were surveyed on non-sunny days. The gender distribution was fairly even, with 266 men (47.58%) and 293 women (52.42%) responding. Age distribution indicated that the majority of respondents were aged between 20 and 39 years, representing 41.68% of the total sample. Those aged 40 to 64 years comprised 28.26%. Information on 15-minute activity, visit frequency, time spent, and social accompaniment is presented in the Appendix. The chi-square and Kruskal–Wallis tests showed no significant differences in age, activity, or clothing between the two parks. However, East Park had older visitors ($\geq$ 40), while Nagai Park had younger visitors.

Environmental parameters

We compared the main thermal environmental parameters at different observation points under similar clear weather conditions (table 2). The solar radiation in the shaded areas (C, A, F) under the tree canopy was less than 166 W/m², which is significantly lower than the levels observed in the open spaces (D, E, B, G). Notably, in the hard-paved areas (D and E), the daily average radiation exceeded 600 W/m². Black globe temperature, which takes into account the effects of air temperature, radiant temperature, and wind speed, is a crucial indicator for assessing the impact of radiative heat on the human body, particularly in outdoor environments. Similar to solar radiation, open spaces exposed to sunlight (D and E) had the highest average daily black globe temperature, exceeding 42°C, while shaded areas (A and C) remained lower and more stable, below 35°C. This highlights the importance of canopy shading in improving thermal comfort. A similar trend was observed in the air temperature, with points D and B exceeding 34°C, while shaded areas A and C remained around 33°C.

In Nagai Park, the daily average temperature at point D was the highest at 35.49°C, followed by point B at 34.22°C. By contrast, the temperatures in shaded areas (points A and C) were the lowest, at 33.57°C and 33.05°C, respectively, indicating that the canopy coverage in these shaded areas played a crucial role in regulating air temperature and enhancing thermal comfort. In East Park, point F had the highest daily average temperature, while the lowest was observed in the lawn area. This could be attributed to the significant cooling effect of the grass, which improved surface temperatures and consequently lowered local air temperatures. RH exhibited a distinct daily variation, with higher levels in the early morning, followed by a gradual decrease over time, especially in open spaces (point D), where it dropped to around 42% at noon. In shaded areas like A and C, humidity remained high and stable, above 51%. This shows that canopy cover helps maintain higher humidity, improving thermal comfort under sunlight.

ARTICLE IN PRESS



ARTICLE IN PRESS

Figure 2. Distribution of Sound Types and Thermal Experience Votes

Acoustic characteristics and types

As shown in the distribution of perceived intensity of sound types across different observation points (Figure 2a), at points B and A, respondents perceived natural sounds most strongly, at 51.43% and 37.97%, respectively, while perceptions of artificial and mechanical sounds were weaker, at 62.86% and 8.57% at B, and 51.90% and 15.19% at A. Point D exhibited the weakest perception of natural sounds, at just 3.57%, but the strongest perception of mechanical sounds, with 78.57% of respondents rating them as strong. At Point C, respondents primarily perceived natural sounds as neutral (61.19%), artificial sounds as weak (44.78%), and mechanical sounds as neutral (44.78%). At Points E and G, respondents predominantly had neutral perceptions of all three sound types. At Point F, respondents primarily perceived natural sounds as neutral (51.79%), while mechanical sounds were perceived as strong (54.46%). Natural sounds dominated in areas beneath the tree canopy and on the botanical garden islands, with minimal mechanical and artificial noise. The entrance square and points E, F, and G were mainly affected by mechanical noise, with weak natural and artificial sounds. Point C, near the main road, experiences both natural and mechanical sounds. Overall, artificial sounds were perceived neutrally.

Distribution of TSVs, TCVs, and TAVs

We calculated the proportion of each vote level for TCVs, TSVs, and TAVs at each observation point (figure 2b). At point D, 89.29% of respondents reported a "$TSV > 0$" (feeling heat), with point D also having the highest proportion (42%) of respondents feeling "very hot." Points C and E followed closely at 35.8% and 32.09%, respectively. Around 10% of respondents

ARTICLE IN PRESS

at points B, C, and D reported a neutral thermal comfort ($TCV = 0$), while 27.68% did so at point F. Discomfort ($TCV < 0$) was highest at point D (78.57%), followed by points B and C (over 70%). More than 50% of respondents rated the environment as "acceptable" ($TAV > 0$) across all observation points. Point A had the highest proportion (83.54%), while point D had the lowest (53.57%). Points C and D also had the highest rates of "unacceptable" votes (around 40%), while point A had the lowest at 8.86%. These results suggest that forest understory areas reduce heat perception, enhancing comfort, while paved surfaces have the opposite effect. The botanical garden's understory received the highest acceptability, while other areas had lower ratings than East Park.

Thermal-Acoustic-Aesthetic on Thermal Experience

Individual effects of thermal, acoustic, and aesthetic factors on thermal experience

We computed the mean TSV (MTSV), mean TCV (MTCV), and mean TAV (MTAV) for each 1°C PET interval bin, fitting them with linear regression models.

(1) Thermal Factor's Effect on Thermal Perception As PET increased, respondents' TSV exhibited an upward trend, while the level of thermal comfort demonstrated a decreasing trend (Figure 3a). The slope of the linear regression equation between MTSV and PET was 0.081, which corresponds to 16.35°C PET/TSV. A neutral temperature is defined as the thermal condition at which individuals perceive neither warmth nor coolness[47]. The neutral physiologically equivalent temperature (NPET) was calculated to be approximately 20.29°C by inputting MTSV = 0. The neutral physiologically equivalent temperature range (NPETR) represents the temperature range corresponding to a TSV of -0.5 to 0.5[50,51]. The NPETR for people was 14.16-–26.51°C.

The linear regression model (Figure 3a) demonstrated that PET was 27.28°C when MTCV was 0, indicating that respondents perceived the thermal environment as "moderate" at this PET level (MTCV = 0, PET = 27.28°C). TCV was significantly negatively correlated with TSV ($p<0.001$). The linear regression analysis (Figure 3b) indicated that when TCV was neutral (TCV = 0), TSV was approximately 0.49. This suggests that respondents favored a slightly warmer environment and might have adapted to it within the neutral range. Thus, when TSV was less than 0.49, respondents were likely to feel comfortable about the thermal environment, and vice versa.

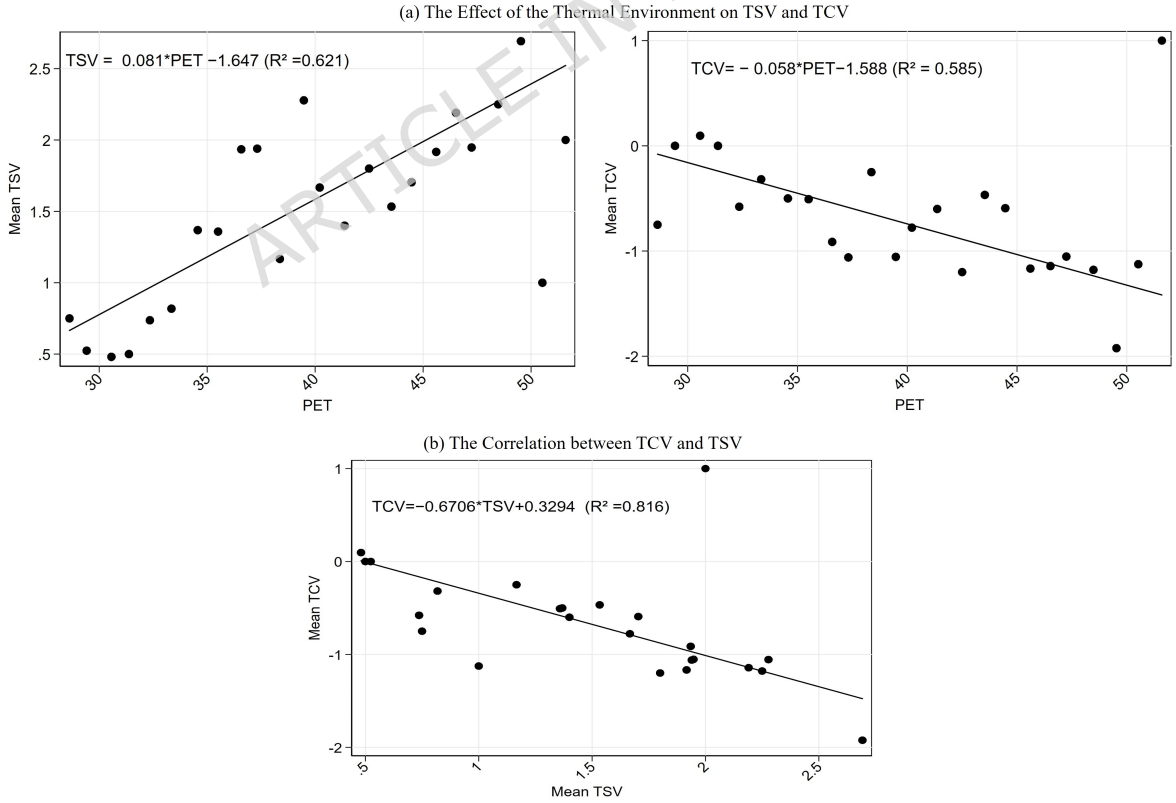


Figure 3. Effect of PET and Thermal Sensation on Thermal Comfort

ARTICLE IN PRESS

(2) Acoustic Factor's Effect on Thermal Experience The correlation analysis (Appendix) showed that LAeq was not significantly correlated with TSV, TCV, or TAV, whereas AcCV was significantly and positively correlated with TCV ($p < 0.05$). The regression results showed that AcCV had a positive effect on TCV, whereas the AcCV×PET interaction term was significantly negative (Table 3), indicating that as PET increases, the positive effect of AcCV on TCV gradually weakens and even disappear, such that acoustic comfort alone is insufficient to offset the adverse experience associated with high heat-stress conditions. In addition, we conducted a sensitivity analysis by varying the PET cutoff in 1°C increments and re-estimating the regression separately for the cool-side (PET $\leq$ cutoff) and hot-side (PET $>$ cutoff) subsamples. The results showed that at lower cutoffs (around 31–34°C), the AcCV coefficient was significantly positive. As the cutoff increased, the coefficient declined overall and was generally negative on the hot side, reaching statistical significance at multiple cutoffs. However, when the cutoff exceeded 47°C, the coefficients on both sides were no longer significant (Figure 4a). Model fit (adjusted R^2) varied with the cutoff: it was relatively higher on the cool side at some lower cutoffs and generally lower on the hot side across most cutoffs (Figure 4b). Overall, these results suggest that acoustic comfort can modestly alleviate heat discomfort under relatively mild thermal conditions, whereas its moderating effect becomes negligible under stronger heat stress.

Table 3. Effect of AcCV on TCV

TCV	Coef.	Std. Err.	p	95% CI Lower	95% CI Upper
AcCV	0.633	0.292	0.031**	0.059	1.207
PET	-0.049	0.004	0.000***	-0.057	-0.042
AcCV×PET	-0.021	0.008	0.005***	-0.036	-0.006
Constant	1.326	0.145	0.000***	1.041	1.612
R-squared		0.603	Number of obs		559
F-test		281.033	Prob > F		0.000

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

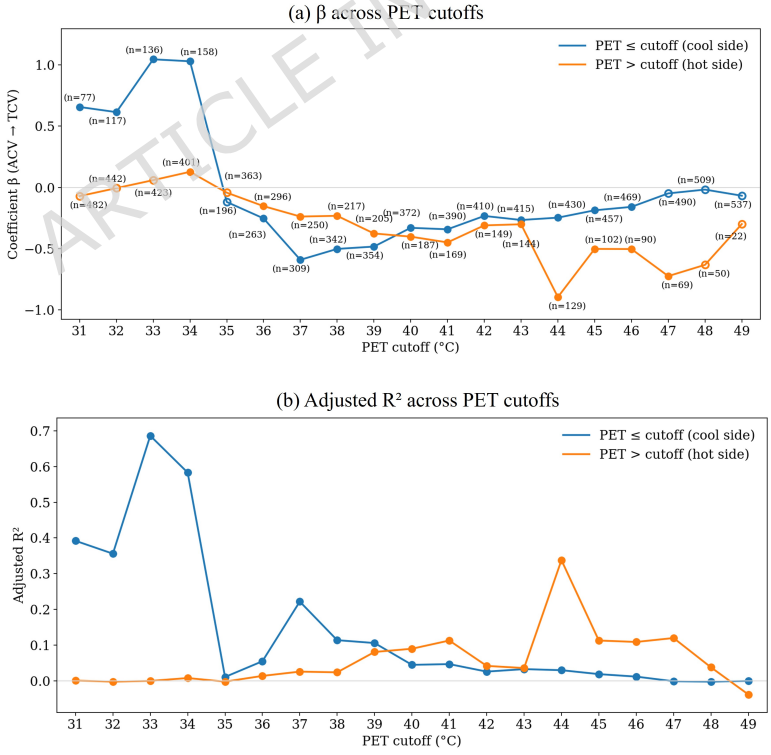


Figure 4. Sensitivity analysis across PET cutoffs. Note: Filled markers indicate statistical significance at $p < 0.05$, whereas hollow markers indicate non-significance for β (AcCV→TCV). n denotes the number of observations in each PET subgroup.

ARTICLE IN PRESS

(3) Aesthetic Factor's Effect on Thermal Experience The linear regression results between TSV, TCV, and AeSV indicate that as aesthetic satisfaction increases, respondents' TSV decreases while TCV increases (Figure 5). Additionally, the low R^2 value of the linear model indicates that although the aesthetic characteristics of the landscape are statistically significant, their explanatory power for respondents' thermal perception is limited compared with thermal conditions. Furthermore, the distribution of aesthetic satisfaction was mainly concentrated between 1 and 2. Therefore, both PET and AeSV were incorporated into the linear regression to further evaluate the model's suitability and the interaction between thermal and aesthetic environment.

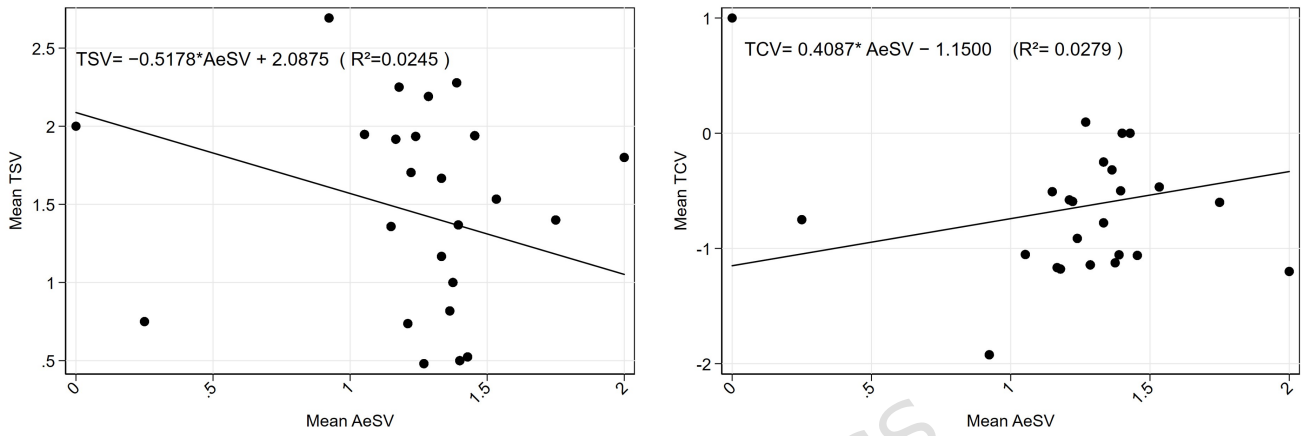


Figure 5. Effects of Aesthetic Satisfaction on Thermal Experience

Interaction effects of thermal, acoustic, and aesthetic factors on thermal experience

Thermal conditions are generally regarded as an important factor influencing thermal perception, while the previous individual-effect analyses indicated that acoustic and aesthetic factors also showed potential associations with thermal experience. Therefore, further exploration of the interactions between these factors and thermal conditions is necessary.Prior to the logistic regression analyses, multicollinearity diagnostics were conducted for the independent variables included in each model. The results indicated that all VIF values were below 5 and the mean VIF values were below 2 (Appendix), suggesting that no significant multicollinearity was present in the models. In addition, for all ordinal regression models, we assessed the proportional odds assumption using Brant tests. In all cases, these tests were non-significant (all $p > 0.05$), indicating that the proportional odds assumption was reasonably satisfied (see Appendix).

(1) Interaction Effects of Thermal and Acoustic Factors on Thermal Perception This study selected LAeq, AcSV, AcCV, and the dominant sound type as the primary indicators of the acoustic environment. Given the significant negative correlation between AcSV and AcCV, which are both based on subjective perception, they were incorporated separately into the ordered logistic regression model to better distinguish their individual effects in interaction with the thermal environment. Meanwhile, LAeq and the dominant sound type were included in the model as key indicators of the acoustic environment. Park type, gender, activity status, and visitation characteristics were added as control variables to account for individual differences. This section's regression results primarily highlight the direction and significance of variable effects. Detailed tables of coefficient estimates and statistical metrics are provided in the Appendix.

PET exerts a highly significant effect on thermal sensation (Table 4): it positively influences TSV, negatively affects TCV and TAV, which is consistent with its independent effect. LAeq and AcSV show no significant effect on thermal perception, whereas AcCV exerts a highly significant positive effect on thermal comfort, which is consistent with its independent effect. The interaction between AcCV and PET suggests that while enhancing acoustic comfort could alleviate thermal discomfort, it might not fully offset the adverse impacts of thermal conditions. Regarding sound type, the dominant sound type shows no statistically significant correlation with TSV and TCV but has a certain impact on TAV. Specifically, compared to an environment without a dominant sound, thermal acceptance is higher when artificial or natural sounds prevail, and lower when mechanical sounds dominate. Regarding individual characteristics, compared to the younger group (under 20 years), respondents aged 20-–39 and 40-–64 years reported lower thermal comfort scores, indicating greater heat sensitivity among middle-aged and older adults. Activity status showed that those standing or walking had higher TSV but lower TCV and TAV, suggesting that moderate activity exacerbates heat discomfort. Respondents with a higher visit frequency (4-–5 times) exhibit greater thermal perception, possibly related to their outdoor activity habits.

ARTICLE IN PRESS

Table 4. Interaction of Thermal and Acoustic Factors on Thermal Experience

(a) Interaction between PET and AcSV				(b) Interaction between PET and AcCV			
Variable	TSV	TCV	TAV	Variable	TSV	TCV	TAV
PET	(+)***	(-)***	(-)***	PET	(+)***	(-)***	(-)***
AcSV				AcCV		(+)***	
PET × AcSV				PET × AcCV		(-)**	
LAeq				LAeq			
Artificial Sound			(+)**	Artificial Sound	(-)*		(+)**
Mechanical Sound			(-)*	Mechanical Sound			(-)*
Natural Sound			(+)**	Natural Sound			(+)*
Park				Park			
Gender				Gender			
Age (20–39)		(-)**		Age (20–39)		(-)**	
Age (40–64)		(-)*		Age (40–64)		(-)*	(-)*
State (standing)	(+)**	(-)*	(-)**	State (standing)	(+)**	(-)*	(-)**
State (walking)	(+)**	(-)*	(-)***	State (walking)	(+)**	(-)*	(-)***
Frequency (4–5 times)	(+)*			Frequency (4–5 times)	(+)*		
Time Spent				Time Spent			
Accompany				Accompany			

Note: *** $p < .01$, ** $p < .05$, * $p < .1$. (+) indicates a positive effect, and (–) indicates a negative effect. Cells left blank denote non-significant effect. Reference groups: dominant sound = none, park = Nagai Park, age = under 20 years, gender = male, state = sitting, frequency = ≤1 time, time spent = < 10 minutes, Accompany = alone.

(2) Interactions of Thermal and Aesthetic Factors on Thermal Experience In this study, AeSV, AeSVG, and AeSVB are all subjective aesthetic evaluation indicators. Because of their strong positive correlation and incomplete measurement of AeSVB at some observation points without water bodies, the AeSVB data are considered incomplete. To avoid potential collinearity issues affecting model estimation and to separately explore the differential effects of various types of aesthetic satisfaction on thermal perception, this section adopted a stratified modeling approach, incorporating each of the three aesthetic satisfaction indicators separately into the ordinal logistic regression model. The regression results in this section primarily report the direction and significance of variable effects. Detailed coefficient estimates and statistical metrics are provided in the Appendix.

The regression results (Table 5) align with the independent effect analysis, confirming that PET is the most stable and direct dominant factor influencing thermal sensation (TSV, TCV, and TAV). Regarding subjective aesthetic satisfaction, neither AeSV nor AeSVG significantly affect TSV or TCV. However, both exhibit a consistent positive effect on TAV, suggesting that higher aesthetic satisfaction enhances respondents' positive acceptance (Tables 5a and 5b). By contrast, although water body aesthetic satisfaction (AeSVB) positively influences TCV and TAV, its interaction effect with PET is negative, indicating that its ability to alleviate heat discomfort is significantly weakened under high heat load conditions (Table 5c). The positive effects of AeSVG are not influenced by PET conditions, whereas those of AeSVB are constrained by the intensity of the thermal environment. Comparatively, aesthetic satisfaction with green spaces exhibits greater stability in enhancing thermal perception. This difference may be attributed to the heterogeneity of water bodies (e.g., fountains, artificial ponds, and natural water bodies) and the relatively small sample size at the corresponding observation points. Overall, green landscapes consistently and robustly enhance subjective thermal perception. This stability may benefit from the natural regulatory functions of green spaces, such as shading, vegetation transpiration, and air circulation, which contribute to enhanced overall thermal comfort and thermal acceptability. Regarding individual attributes, the results largely align with previous analyses. Additionally, respondents who spent a relatively long travel time (40-–60 minutes) to get to the park reported lower thermal acceptability. By contrast, those accompanied by family members demonstrated higher thermal comfort and acceptability compared to respondents traveling alone.

(3) Effects of the Thermal-Acoustic-Aesthetic Factors on Overall Satisfaction Controlling for potential confounders, such as individual attributes and access characteristics, this study employed an ordered logistic regression model to examine the effects of PET, acoustic comfort, and overall aesthetic satisfaction on respondents' overall satisfaction. For clarity, only statistically significant variables and their corresponding coefficients are reported in this section.

The results (Table 6) show that PET has a significant negative effect on overall satisfaction, suggesting that higher thermal

ARTICLE IN PRESS

Table 5. Interaction of Thermal and Aesthetic Factors on Thermal Experience

(a) Interaction between PET and AeSV				(b) Interaction between PET and AeSVG				(c) Interaction between PET and AeSVB			
	TSV	TCV	TAV		TSV	TCV	TAV		TSV	TCV	TAV
PET	(+)***	(-)***	(-)***	PET	(+)***	(-)***	(-)***	PET	(+)***	(-)**	(-)***
AeSV			(+)**	AeSVG			(+)**	AeSVB		(+)***	(+)**
PET*AeSV				PET*AeSVG				PET*AeSVB		(-)***	(-)**
Park				park				Park (East park)	(+)**	(-)*	
Gender				gender				Gender			(-)*
Age (20-39)		(-)**		Age (20-39)		(-)**		Age (20-39)	(-)**		
Age (40-64)		(-)**	(-)*	Age (40-64)		(-)**	(-)*	Age (40-64)			(-)*
State (sitting)	(+)**	(-)*	(-)**	State (sitting)	(+)**	(-)**	(-)**	State (Light exercise)	(-)**		
State (walking)	(+)**	(-)**	(-)***	State (walking)	(+)**	(-)**	(-)***	Time spent (20-30 mins)			(-)**
Frequency (4-5 times)	(+)**			Frequency (4-5 times)	(+)*			Time spent (30-60 mins)			(-)**
Time Spent				Time Spent							
Accompany				Accompany				Accompany (Family)		(+)**	(+)**

Note: *** $p < .01$, ** $p < .05$, * $p < .1$. (+) indicates a positive effect, and (–) indicates a negative effect. Cells left blank denote non-significant effect. Reference groups: dominant sound = none, park = Nagai Park, age = under 20 years, gender = male, state = sitting, frequency = ≤1 time, time spent = < 10 minutes, Accompany = alone.

ARTICLE IN PRESS

Table 6. Effect of Thermal, Acoustic, and Aesthetic Factors on Overall Satisfaction

						95% Confidence Interval	
OSV	Coef.	St.Err.	z	p	Odds Ratio	Lower	Upper
PET	-0.032	0.014	-2.220	0.026**	0.968	0.941	0.996
TCV							
AcCV	0.274	0.082	3.330	0.001***	1.316	1.120	1.546
AeSV	1.664	0.112	14.840	0.000***	5.281	4.239	6.579
Park							
Gender							
Age (65+)	0.634	0.350	1.810	0.070*	1.886	0.951	3.742
State							
Frequency							
Time Spent							
Accompany							
Pseudo r-squared		0.228		Number of obs	545		
Chi-square		355.151		Prob > chi2	0.000		
Akaike crit. (AIC)		1257.530		Bayesian crit. (BIC)	1377.952		

Note: *** $p < .01$, ** $p < .05$, * $p < .1$. (+) indicates a positive effect, and (–) indicates a negative effect. Cells left blank denote non-significant effect. Reference groups: dominant sound = none, park = Nagai Park, age = under 20 years, gender = male, state = sitting, frequency = ≤1 time, time spent = < 10 minutes, Accompany = alone.

ARTICLE IN PRESS

load is associated with lower satisfaction. By contrast, both acoustic satisfaction and overall aesthetic satisfaction have significant positive effects, with the influence of aesthetic satisfaction being particularly pronounced. Thermal comfort is not statistically significant in the model, possibly owing to its strong negative correlation with PET. The dominant effect of PET may overshadow the independent contribution of TCV. Furthermore, respondents aged 65 years and above reported higher levels of overall satisfaction. By contrast, gender, park type, activity status, and visit characteristics show no statistically significant effects on overall satisfaction. These findings highlight the importance of aesthetic quality, acoustic comfort, and thermal conditions in improving outdoor environmental comfort.

Discussion

Understanding thermal comfort in urban parks is crucial for enhancing the overall quality of public spaces. While microclimatic conditions are widely recognized as the primary factors influencing thermal comfort, this study extends the focus to explore how other environmental factors, such as acoustics and aesthetics, and their interactions with thermal conditions, affect thermal comfort. In this section, we discuss the key findings of this study and their implications for improving urban park environments.

Dominance of Thermal Conditions in Thermal Experience

Thermal conditions dominate residents' thermal experiences, consistent with previous studies[20,52]. Human thermal perception primarily occurs through the thermal receptors in the skin, which detect changes in the surrounding environmental temperature and transmit signals to the brain via the nervous system. Excessive heat or cold directly affects the body's comfort, health, and even physiological functions, making the dominant role of temperature in thermal perception particularly evident. Our study found that an increase in PET heightens perceived heat levels, while reducing thermal comfort and acceptance. This finding is consistent across numerous studies. However, neutral PET varies across different climatic regions. In this study, the Cfa-based neutral PET is found to be 20.29°C, slightly lower than those reported in other studies in Cfa areas[53–55]. Although the cities in these studies are all located in the Cfa climate zone, their summer maximum temperatures are generally higher than those in the region of this study. The climatic differences, particularly higher temperatures, likely contribute to the variation in neutral PET values, as elevated temperatures typically lead to higher neutral PET. Differences in respondents' thermal experience and tolerance may further influence the results. Additionally, in terms of individual characteristics and park visitation patterns, respondents in the middle and older age groups (20-64 years) reported lower TCV and lower TAV than the younger group did. During moderate-intensity activities, such as standing or walking, TSV increased, while both thermal comfort and acceptability decreased. Respondents who were accompanied by family members reported higher levels of thermal comfort and acceptance than those who visited the park alone.

Multi-sensory Modulation of Thermal Experience Is Influenced by Thermal Condition

Our study found that LAeq and AcSV were not significantly correlated with TSV or TCV, which is consistent with the results of Mohammadzadeh[56].However, previous research suggests that the effect of LAeq on thermal comfort may be seasonally or context dependent[20]. In contrast, AcCV showed a weak positive association with TCV, but this relationship was constrained by PET and was no longer significant at higher PET levels. This suggests that under excessively hot conditions, the compensatory role of acoustic comfort in improving thermal comfort is limited. Additionally, perceived natural sound types were significantly associated with higher thermal acceptability, whereas mechanical noise showed the opposite pattern, which is consistent with the findings of Chen[57].Natural sounds (e.g., birdsong and flowing water) have been shown to reduce stress, promote psychological restoration, and enhance attention[58,59].Moreover, emotions mediate the effects of landscape elements on thermal comfort, with pleasant emotions having a particularly strong influence on both thermal comfort and thermal acceptability[44].In addition, sound types may serve as proxies for environmental conditions. Natural sounds are often present in spaces with high vegetation cover, ample shading, and significant evapotranspiration, which can improve the microclimate, reduce heat stress, and enhance thermal comfort. In contrast, mechanical noise is typically found in areas with heavy traffic, extensive impervious surfaces, and pronounced urban heat island effects, where outdoor thermal conditions are generally less favorable. Thus, sound types may also reflect broader urban and microclimatic characteristics. Whether the observed effects arise primarily from cross-modal influences of sound itself or from its role as an environmental proxy, and how these pathways interact, requires further study through more rigorous experimental designs (such as laboratory studies or field experiments with controlled environmental variables) as well as longitudinal studies (for example, long-term tracking of thermal perceptions under different environmental conditions). Such research would help disentangle psychological effects from environmental proxy effects and clarify their causal mechanisms

Our study shows that aesthetic satisfaction helps reduce thermal sensation and enhancing thermal comfort and acceptance, supporting previous findings. Groups with higher satisfaction with acoustic and aesthetic perceptions reported significantly higher thermal comfort votes[56]. Similar findings were observed in a study focusing on older adults, where TSV and TCV were influenced by both the acoustic and visual environment[60]. Visual connection to nature has been demonstrated to have

ARTICLE IN PRESS

a positive impact on attention restoration, stress reduction, and overall health and well-being. Stress reduction theory posits that exposure to nature promotes stress recovery[61]. Aesthetically pleasing landscapes help people relieve the stress and strain of their living environment. Natural sceneries can help residents reduce stress, anxiety, and tension[40], while also increasing thermal acceptability and thermal comfort[41]. The affect-as-information theory suggests that individuals use their current emotional state as a cue for evaluating the external environment. Prolonged exposure to aesthetically valuable blue-green spaces may enhance individuals' emotional attachment and sense of identification with these spaces. This attachment further increases their tolerance and satisfaction with the quality of the environment. Even when objective conditions are suboptimal (e.g., higher temperatures, higher noise levels), individuals may still provide a positive evaluation of the environment owing to the combined influence of aesthetic and emotional factors. This interaction between bottom–up perceptual mechanisms and top–down psychological regulation mechanisms plays a crucial role in shaping environmental assessments. In addition, the visual landscape has a spiritual impact on people's health and thermal comfort[31,62,63]. Furthermore, through the analysis of its interaction with the thermal environment, our study found that although AeSV positively influences thermal comfort, its effect is insufficient to offset the dominant impact of thermal stress. Notably, the coefficient of the interaction term is significantly lower than that of AeSV alone, highlighting the important role of aesthetic factors in enhancing the thermal environment, despite the dominant influence of thermal conditions. Overall, although acoustic satisfaction and visual satisfaction have a more limited impact on thermal comfort than the direct effect of the thermal environment on thermal perception does, they still warrant significant attention. People recognize environments through multi-sensory integration. The multisensory integration theory suggests that the perception of visual stimuli, such as greenery and water surfaces, along with natural sounds in the environment, can create expectations of a cool and comfortable space. These expectations, in turn, influence individuals' actual perception and tolerance of the thermal environment through emotional and psychological regulation.

Aesthetic Satisfaction for Green Spaces Enhances Thermal Comfort More Than for Blue Spaces

Numerous studies have demonstrated that visual comfort can enhance OTC to some extent, as previously discussed. However, research focusing on the distinct effects of aesthetic satisfaction with blue and green spaces on thermal comfort remains limited. Based on the above, this study provides a separate analysis of aesthetic satisfaction with blue and green spaces, a perspective that has received limited attention in existing literature. Our study confirmed that aesthetic satisfaction with green spaces had a similar impact on thermal perception as overall satisfaction. Our results show that higher aesthetic satisfaction with green spaces is associated with lower thermal perception, higher thermal comfort, and thermal acceptability. Its interaction with thermal conditions has no significant effect on TSV, TCV, and TAV. However, the interaction between aesthetic satisfaction with blue spaces and thermal conditions shows a different pattern: it negatively affects TCV and TAV. This result suggests that the aesthetic satisfaction with green spaces contributes more consistently to thermal acceptability than does that with blue spaces, and its positive effect is not limited or moderated by thermal conditions. The underlying mechanism may be related to the different ways in which blue and green spaces mitigate heat. Green spaces, particularly vegetation, have been shown to mitigate heat effects in numerous studies[64]. Our study found that stable understory spaces are the most effective cooling areas, significantly reducing radiation and temperature. Furthermore, we measured water temperatures under various conditions and found that the temperature of flowing water under tree shade was significantly lower than that of still water exposed to direct sunlight. Gunawardena et al. confirmed that tree-dominated green spaces provide greater heat stress relief when most needed[65]. The presence of smaller, scattered water bodies does not significantly improve thermal comfort compared to a single large water body. Research has indicated that although the total area of multiple small water bodies may exceed that of one large water body, their cooling effect is still less effective[66]. Research also suggests that, while water bodies without trees may reduce air temperature through evaporation, they can simultaneously increase humidity, which diminishes their beneficial effects on thermal comfort. By contrast, combining water bodies with trees yields superior outcomes in regulating urban micro-climates and enhancing thermal comfort[67]. Moreover, the effectiveness of water bodies often depends on additional factors, such as wind speed and surrounding vegetation, with both stagnant and flowing water influencing water temperature[65]. Green spaces enhance cooling effects not only through physiological mechanisms, such as shading and transpiration, but also through visual perception and emotional regulation. By contrast, although blue spaces are associated with a sense of coolness, their physiological cooling mechanisms are inconsistent. Especially in high-temperature environments, when the temperature of water bodies is higher, the temperature perception effect of the blue visual cue may be diminished.

Balancing Thermal, Acoustic, and Aesthetic Environments for a Better Outdoor Experience

Improving park comfort significantly enhances residents' well-being, particularly during hot summers when thermal comfort becomes crucial. Our study found a significant negative association between higher PET levels and overall satisfaction. By contrast, improvements in TCV, AcCV, and AeSV contributed to notable enhancements in comfort. This highlights the importance of integrating thermal, acoustic, and aesthetic factors, not only to improve thermal comfort but also to enhance the overall sensory experience in urban parks. Multisensory interaction is becoming a key focus in future urban park design and management. It incorporates not only visual and auditory elements, such as landscapes and soundscapes, but also olfactory

ARTICLE IN PRESS

and tactile experiences to create more engaging and restorative environments. Previous research has shown that in urban microgreen spaces with relatively low ambient noise, multisensory combinations—such as visual–auditory, visual–olfactory, and visual–auditory–olfactory stimuli—are more effective in restoring visual attention than visual stimuli alone. Furthermore, integrating these sensory modalities enhances spatial perception by leveraging positive sensory inputs to mask or alleviate negative environmental experiences[68]. Participants have shown a preference for visually biodiverse environments. However, overly diverse auditory and olfactory stimuli could lead to sensory overload and discomfort[69]. Therefore, balancing multisensory interaction and thermal comfort requires considering the interplay between various environmental factors and thermal conditions. Achieving effective cooling requires carefully weighing these interactions. Additionally, to address distinct thermal needs in winter and summer, design strategies could include the strategic selection of evergreen and deciduous tree species, as well as the use of temporary artificial structures. This study emphasizes the close relationship between thermal comfort and the multisensory integration of thermal, acoustic, and aesthetic factors, offering valuable insights for developing multisensory design strategies in future urban park planning.

Optimizing Soundscape Measurement in Urban Parks

In typical urban park open spaces, the acoustic environment—a key factor influencing public experience—not only affects psychological responses but also interacts with thermal perception. This study measured the acoustic environment objectively using the LAeq, and subjectively through sound intensity perception (AcSV), satisfaction ratings, and perceived intensity of different sound types. The results show no significant correlation between LAeq or AcSV and thermal perception, although AcCV has a significant positive effect on thermal comfort evaluation (TCV). Possible reasons for this result include the following. First, although LAeq varies across observation points, overall noise levels are concentrated between 50 and 70 dBA with limited fluctuation, which may be insufficient to significantly affect thermal perception. Second, as an energy-averaging metric, LAeq ignores the spectral structure and perceptual characteristics of sound, limiting its ability to capture subjective human sound experience. AcSV reflects respondents' perception of overall sound intensity but lacks detail on sound types, frequency distribution, and psychoacoustic effects.

Psychoacoustic metrics are widely used to address limitations of single intensity measures in acoustics and psychology. For example, loudness quantifies subjective sound intensity perception, aligning more closely with human experience than sound pressure level. Sharpness indicates the proportion of high-frequency components, with sharper sounds perceived as more piercing. Roughness and fluctuation strength characterize fast and slow temporal modulations, often linked to tension and discomfort. These metrics are calculated through spectral analysis of raw sound signals, commonly using fast Fourier transform for frequency-domain processing.

Although this study examined the effects of acoustics and thermal environment, no on-site sound environment samples were collected owing to the experimental design and logistical constraints. Consequently, it was not feasible to extract psychoacoustic features from the original audio or perform spectral analysis, limiting a deeper understanding of the sound-–thermal sensation interaction mechanism. Current research has primarily focused on public preferences for different sound types and acoustic environment evaluations (e.g., preference for natural sounds)[70,71], while studies integrating spectral characteristics to analyze thermal perception remain relatively scarce. Different sound types (e.g., natural, artificial, and mechanical) exhibit characteristic frequency spectrum patterns. Natural sounds are generally broadband, low-frequency, and continuous, whereas traffic noise tends to be intermittent with sharp high-frequency components. Future investigations combining sound types and spectral characteristics to understand their influence on thermal comfort may identify novel regulatory mechanisms.

Limitations

While this study provides valuable insights into the relationship between acoustic, aesthetic, and thermal conditions in urban environments, there are some limitations that should be addressed in future research. (1) This study employed Class-2 acoustic measurement equipment, which has certain accuracy limitations compared to Class-1 devices, and did not incorporate psychoacoustic indicators. Therefore, it is difficult to fully capture the impact of the acoustic environment on public thermal perception. Future research should improve both the precision and dimensionality of acoustic measurements, following the systematic methodological framework of the ISO 12913 series, which includes perceptual surveys and audio recordings. A more comprehensive experimental design should be developed by introducing psychoacoustic parameters, such as loudness, roughness, and sharpness, combined with spectral analysis of various sound types, to thoroughly investigate the potential relationships between spectral features and thermal comfort. (2) Richness of data collection: This study not only investigated summer sunny day data, a common focus of most studies, but also included data from non-sunny days. However, owing to weather constraints during the data-collection period, non-sunny day data did not cover all observation points. Future research could incorporate mobile measurement methods, which would extend the research scope, providing a broader dataset on thermal comfort and compensating for the limitations of fixed-point measurements. (3) In-depth exploration of multi-framework theories and mechanisms: This study analyzed the interactive effects of acoustics, aesthetics, and thermal conditions on thermal comfort but did not establish a comprehensive multi-theory framework. Future research should further integrate additional sensory

ARTICLE IN PRESS

factors, such as smell and touch, while combining physiological indicators (e.g., heart rate, cortisol levels) and psychological measures (e.g., mood scores) to develop a more systematic, multi-sensory, interdisciplinary, and multi-theoretical research framework, enabling a more thorough investigation of their underlying mechanisms.

Conclusions

Based on multiple weather conditions and various types of parks, this study systematically explored the interactive effects of thermal, acoustic, and aesthetic environments on outdoor thermal experience in urban parks from the perspective of multi-sensory environmental perception, with a particular emphasis on the dominant role of the thermal environment in thermal perception. Acoustic and aesthetic comfort may help alleviate thermal discomfort to some extent, but their overall moderating effects are limited, particularly for acoustic comfort, whose positive influence tends to disappear under strong thermal stress.Moreover, acoustic and aesthetic comfort exerted a significantly greater influence on overall environmental satisfaction than thermal comfort. This study not only highlights the importance of thermal conditions, but also proposes an innovative design framework for multi-sensory collaborative optimization—integrating thermal, acoustic, visual, olfactory, and tactile dimensions—which would provide both theoretical insights and practical guidance for enhancing thermal comfort and multi-sensory quality in urban open spaces.

Data availability

Data will be made available upon reasonable request by contacting the corresponding author.

References

1. Tuholske, C. *et al.* Global urban population exposure to extreme heat. *Proc. Natl. Acad. Sci.* **118**, e2024792118 (2021).
2. Watkins, R., Palmer, J. & Kolokotroni, M. Increased temperature and intensification of the urban heat island: Implications for human comfort and urban design. *Built Environ.* **33**, 85–96 (2007).
3. Ahmed, K. S. Comfort in urban spaces: defining the boundaries of outdoor thermal comfort for the tropical urban environments. *Energy buildings* **35**, 103–110 (2003).
4. Loures, L., Santos, R. & Panagopoulos, T. Urban parks and sustainable city planning-the case of portimão, portugal. *population* **15**, 171–180 (2007).
5. Braubach, M. *et al.* Effects of urban green space on environmental health, equity and resilience. *Nature-based solutions to climate change adaptation urban areas: linkages between science, policy practice* 187–205 (2017).
6. Kumar, P. & Sharma, A. Assessing the outdoor thermal comfort conditions of exercising people in the semi-arid region of india. *Sustain. Cities Soc.* **76**, 103366 (2022).
7. Yahia, M. W. & Johansson, E. Evaluating the behaviour of different thermal indices by investigating various outdoor urban environments in the hot dry city of damascus, syria. *Int. journal biometeorology* **57**, 615–630 (2013).
8. Zhang, Y., Rao, F., Xue, J. & Lai, D. Dependence of urban park visits on thermal environment and air quality. *Urban For. & Urban Green.* **79**, 127813 (2023).
9. De Dear, R. & Brager, G. S. Developing an adaptive model of thermal comfort and preference. *UC Berkeley: Cent. for Built Environ.* (1998).
10. Wu, H., Sun, X. & Wu, Y. Investigation of the relationships between thermal, acoustic, illuminous environments and human perceptions. *J. Build. Eng.* **32**, 101839 (2020).
11. Schweiker, M. *et al.* Review of multi-domain approaches to indoor environmental perception and behaviour. *Build. Environ.* **176**, 106804 (2020).
12. Torresin, S., Pernigotto, G., Cappelletti, F. & Gasparella, A. Combined effects of environmental factors on human perception and objective performance: A review of experimental laboratory works. *Indoor air* **28**, 525–538 (2018).
13. Tsai, K.-T. & Lin, Y.-H. Identification of urban park activity intensity at different thermal environments and visible sky by using sound levels. *Int. journal biometeorology* **62**, 1987–1994 (2018).
14. Galindo, T. & Hermida, M. A. Effects of thermophysiological and non-thermal factors on outdoor thermal perceptions: The tomebamba riverbanks case. *Build. Environ.* **138**, 235–249 (2018).
15. Nagano, K. & Horikoshi, T. New comfort index during combined conditions of moderate low ambient temperature and traffic noise. *Energy Build.* **37**, 287–294 (2005).

ARTICLE IN PRESS

16. Pellerin, N. & Candas, V. Combined effects of temperature and noise on human discomfort. *Physiol. & Behav.* **78**, 99–106 (2003).

17. Pellerin, N. & Candas, V. Effects of steady-state noise and temperature conditions on environmental perception and acceptability. *Indoor air* **14** (2004).

18. Qin, Z. *et al.* Creating comfortable outdoor environments: Understanding the intricate relationship between sound, humidity, and thermal comfort. *Urban Clim.* **55**, 101967 (2024).

19. Lau, K. K.-L. & Choi, C. Y. The influence of perceived aesthetic and acoustic quality on outdoor thermal comfort in urban environment. *Build. Environ.* **206**, 108333 (2021).

20. Jin, Y., Jin, H. & Kang, J. Combined effects of the thermal-acoustic environment on subjective evaluations in urban squares. *Build. Environ.* **168**, 106517 (2020).

21. Fanger, P. O., Breum, N. & Jerking, E. Can colour and noise influence man's thermal comfort? *Ergonomics* **20**, 11–18 (1977).

22. Yang, W., Moon, H. J. & Kim, M.-J. Combined effects of short-term noise exposure and hygrothermal conditions on indoor environmental perceptions. *Indoor Built Environ.* **27**, 1119–1133 (2018).

23. Wen, X. *et al.* Exploring the boundaries of indoor combined thermal-acoustic environmental effects on comfort perceptions. *Build. Environ.* 112580 (2025).

24. Luo, W. *et al.* Effects of correlated color temperature of light on thermal comfort, thermophysiology and cognitive performance. *Build. Environ.* **231**, 109944 (2023).

25. Lechner, S., Moosmann, C., Wagner, A. & Schweiker, M. Does thermal control improve visual satisfaction? interactions between occupants' self-perceived control, visual, thermal, and overall satisfaction. *Indoor Air* **31**, 2329–2349 (2021).

26. Brambilla, A., Hu, W., Samangouei, R., Cadorin, R. & Davis, W. How correlated colour temperature manipulates human thermal perception and comfort. *Build. Environ.* **177**, 106929 (2020).

27. Qin, H. *et al.* The effects of brightness and prominent colors on outdoor thermal perception in chongqing, china. *Int. J. Biometeorol.* **68**, 1143–1154 (2024).

28. Huebner, G. M. *et al.* Saving energy with light? experimental studies assessing the impact of colour temperature on thermal comfort. *Energy Res. & Soc. Sci.* **15**, 45–57 (2016).

29. Candas, V. & Dufour, A. Thermal comfort: multisensory interactions? *J. physiological anthropology applied human science* **24**, 33–36 (2005).

30. Lenzholzer, S. *et al.* Awareness of urban climate adaptation strategies–an international overview. *Urban climate* **34**, 100705 (2020).

31. Lin, T.-P., Tsai, K.-T., Hwang, R.-L. & Matzarakis, A. Quantification of the effect of thermal indices and sky view factor on park attendance. *Landsc. Urban Plan.* **107**, 137–146 (2012).

32. Mayes, H. S. *et al.* The effects of manipulating the visual environment on thermal perception: A structured narrative review. *J. Therm. Biol.* **112**, 103488 (2023).

33. Allan, A. C., Garcia-Hansen, V., Isoardi, G. & Smith, S. S. Subjective assessments of lighting quality: A measurement review. *Leukos* (2019).

34. Pierson, C., Wienold, J. & Bodart, M. Review of factors influencing discomfort glare perception from daylight. *Leukos* **14**, 111–148 (2018).

35. Bourchtein, A., Bourchtein, L. & Naoumova, N. On the visual complexity of built and natural landscapes. *Fractals* **22**, 1450008 (2014).

36. Ode, Å., Hagerhall, C. M. & Sang, N. Analysing visual landscape complexity: theory and application. *Landsc. research* **35**, 111–131 (2010).

37. Sunkavalli, K., Romeiro, F., Matusik, W., Zickler, T. & Pfister, H. What do color changes reveal about an outdoor scene? In *2008 IEEE Conference on Computer Vision and Pattern Recognition*, 1–8 (IEEE, 2008).

38. Kirchhoff, T., Ramisch, K., Feucht, T., Reif, C. & Suda, M. Visual evaluations of wind turbines: Judgments of scenic beauty or of moral desirability? *Landsc. Urban Plan.* **226**, 104509 (2022).

39. Spielhofer, R., Hunziker, M., Kienast, F., Hayek, U. W. & Grêt-Regamey, A. Does rated visual landscape quality match visual features? an analysis for renewable energy landscapes. *Landsc. urban planning* **209**, 104000 (2021).

ARTICLE IN PRESS

ARTICLE IN PRESS

40. Drahota, A. *et al.* Sensory environment on health-related outcomes of hospital patients. *Cochrane Database Syst. Rev.* (2012).

41. Meili, N. *et al.* Vegetation cover and plant-trait effects on outdoor thermal comfort in a tropical city. *Build. environment* **195**, 107733 (2021).

42. Chan, S., Chau, C. K. & Leung, T. On the study of thermal comfort and perceptions of environmental features in urban parks: A structural equation modeling approach. *Build. environment* **122**, 171–183 (2017).

43. Xu, X. *et al.* The cooling and energy saving effect of landscape design parameters of urban park in summer: A case of beijing, china. *Energy Build.* **149**, 91–100 (2017).

44. Yan, T., Jin, H. & Jin, Y. The mediating role of emotion in the effects of landscape elements on thermal comfort: A laboratory study. *Build. Environ.* **233**, 110130 (2023).

45. Hodder, S. G. & Parsons, K. The effects of solar radiation on thermal comfort. *Int. journal biometeorology* **51**, 233–250 (2007).

46. Norge, S. Ergonomics of the thermal environment-instruments for measuring physical quantities. *NS-EN ISO* **7726**, 2001 (2001).

47. Standard, A. Thermal environmental conditions for human occupancy. *ANSI/ASHRAE, 55* **5** (1992).

48. Potchter, O., Cohen, P., Lin, T.-P. & Matzarakis, A. Outdoor human thermal perception in various climates: A comprehensive review of approaches, methods and quantification. *Sci. Total. Environ.* **631**, 390–406 (2018).

49. Shooshtarian, S. & Ridley, I. The effect of physical and psychological environments on the users thermal perceptions of educational urban precincts. *Build. Environ.* **115**, 182–198 (2017).

50. Nikolopoulou, M. & Lykoudis, S. Use of outdoor spaces and microclimate in a mediterranean urban area. *Build. environment* **42**, 3691–3707 (2007).

51. Krüger, E., Drach, P., Emmanuel, R. & Corbella, O. Assessment of daytime outdoor comfort levels in and outside the urban area of glasgow, uk. *Int. journal biometeorology* **57**, 521–533 (2013).

52. Zhen, M., Chen, Z. & Zou, Q. Combined effects of thermal and acoustic environments on outdoor human comfort in university campus. *Urban Clim.* **49**, 101566 (2023).

53. Zeng, Y. & Dong, L. Thermal human biometeorological conditions and subjective thermal sensation in pedestrian streets in chengdu, china. *Int. journal biometeorology* **59**, 99–108 (2015).

54. Fang, Z. *et al.* Investigation of outdoor thermal comfort prediction models in south china: A case study in guangzhou. *Build. Environ.* **188**, 107424 (2021).

55. Xiong, J. *et al.* A study of waterside microenvironmental factors and their effects on summer outdoor thermal comfort in a cfa-climate campus. *J. Therm. Biol.* **117**, 103700 (2023).

56. Mohammadzadeh, N., Karimi, A. & Brown, R. D. The influence of outdoor thermal comfort on acoustic comfort of urban parks based on plant communities. *Build. Environ.* **228**, 109884 (2023).

57. Chen, J., Jin, Y. & Jin, H. Effects of visual landscape on subjective environmental evaluations in the open spaces of a severe cold city. *Front. Psychol.* **13**, 954402 (2022).

58. Ratcliffe, E., Gatersleben, B. & Sowden, P. T. Bird sounds and their contributions to perceived attention restoration and stress recovery. *J. Environ. Psychol.* **36**, 221–228 (2013).

59. Zhu, R. *et al.* Effects of natural sound exposure on health recovery: A systematic review and meta-analysis. *Sci. Total. Environ.* **921**, 171052 (2024).

60. Du, M., Hong, B., Gu, C., Li, Y. & Wang, Y. Multiple effects of visual-acoustic-thermal perceptions on the overall comfort of elderly adults in residential outdoor environments. *Energy Build.* **283**, 112813 (2023).

61. Ulrich, R. S. *et al.* Stress recovery during exposure to natural and urban environments. *J. environmental psychology* **11**, 201–230 (1991).

62. Nikolopoulou, M. & Steemers, K. Thermal comfort and psychological adaptation as a guide for designing urban spaces. *Energy buildings* **35**, 95–101 (2003).

63. Lenzholzer, S. & de Vries, S. Exploring outdoor thermal perception—a revised model. *Int. journal biometeorology* **64**, 293–300 (2020).

ARTICLE IN PRESS

ARTICLE IN PRESS

64. Shashua-Bar, L., Pearlmutter, D. & Erell, E. The influence of trees and grass on outdoor thermal comfort in a hot-arid environment. *Int. journal climatology* **31**, 1498–1506 (2011).

65. Gunawardena, K. R., Wells, M. J. & Kershaw, T. Utilising green and bluespace to mitigate urban heat island intensity. *Sci. Total. Environ.* **584**, 1040–1055 (2017).

66. Theeuwes, N. E., Solcerova, A. & Steeneveld, G. J. Modeling the influence of open water surfaces on the summertime temperature and thermal comfort in the city. *J. Geophys. Res. Atmospheres* **118**, 8881–8896 (2013).

67. Teshnehdel, S., Gatto, E., Li, D. & Brown, R. D. Improving outdoor thermal comfort in a steppe climate: Effect of water and trees in an urban park. *Land* **11**, 431 (2022).

68. Zheng, H., Luo, M., Wang, Y. & Wei, Y. Multi-sensory interaction and spatial perception in urban microgreen spaces: A focus on vision, auditory, and olfaction. *Sustainability* **16**, 8809 (2024).

69. Hao, S. *et al.* Multisensory urban design: Healing effects of visual, auditory, and olfactory enhancements in street landscapes. *Meng, Multisensory Urban Des. Heal. Eff. Vis. Auditory, Olfactory Enhancements Str. Landscapes* .

70. Kang, J. & Zhang, M. Semantic differential analysis of the soundscape in urban open public spaces. *Build. environment* **45**, 150–157 (2010).

71. Jeon, J. Y., Lee, P. J., You, J. & Kang, J. Perceptual assessment of quality of urban soundscapes with combined noise sources and water sounds. *The J. Acoust. Soc. Am.* **127**, 1357–1366 (2010).

Author contributions

Yu Zhang: Conceptualization, Methodology, Data curation, Visualization, Investigation, Writing-Original draft preparation. Yuta Uchiyama: Conceptualization, Methodology, Writing-Review, & Editing. Masayuki Sato: Project administration, Supervision, Conceptualization, Methodology, Writing-Review, & Editing.

Funding

This work was financially supported by Grant-in-Aid for Scientific Research [23K25067]; Environment Research and Technology Development Fund by the Ministry of the Environment, Japan [1FS-2201]; University-Driven Urban Innovation Kobe, Kobe city [A25101]. Open Access funding was provided by Kobe University.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.S.

ARTICLE IN PRESS